

Sleep Health and Lifestyle Analysis

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Agenda

Results of sleep health and lifestyle analysis

- Data Description
- Typical Amount of Physical Activity
- Number of Daily Steps
- Distribution of Heart Rates

Data Description

374 rows • 13 variables • Kaggle “Sleep Health and Lifestyle” dataset

- Demographics: Age (27–59), Gender, Occupation (11 roles, unevenly sampled)
- Lifestyle: Sleep Duration, Physical Activity Level, Daily Steps, Stress Level
- Health: BMI Category, Blood Pressure, Heart Rate, Sleep Disorder
- Data quality: Sleep Disorder blank for 58.6% of rows – recoded as “None” rather than dropped; “Normal” and “Normal Weight” BMI labels merged
- 64.7% of rows are duplicate lifestyle profiles under a new Person ID – effective distinct sample is closer to ~132 than 374

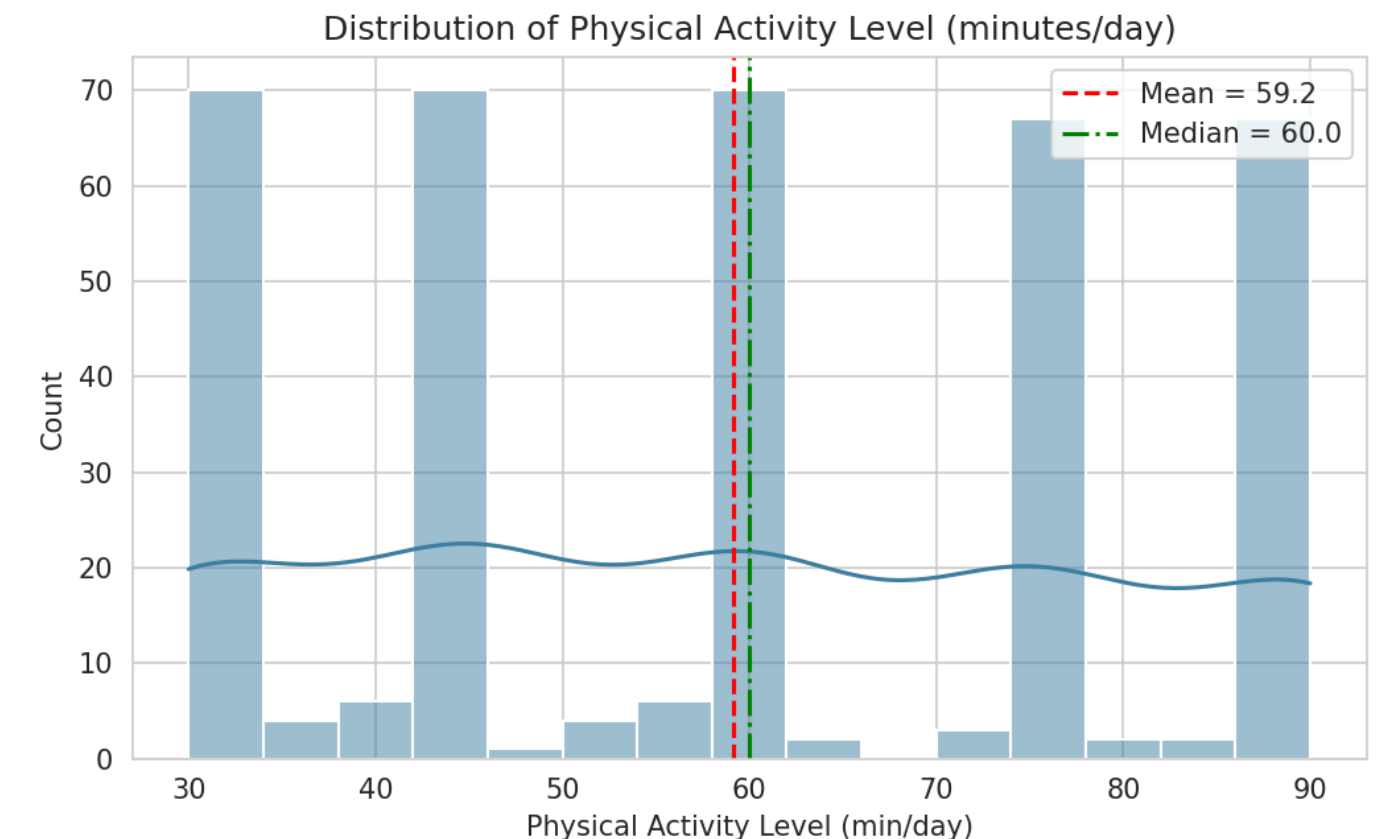
Variable Classifications

Variable	Type	Nominal / Ordinal / Continuous / Discrete	Reasoning
Person ID	Categorical	Nominal	Identifier only — arithmetic on it is meaningless
Gender	Categorical	Nominal	Two unordered labels
Age	Numeric	Continuous	Whole years appears as Discrete, but wide enough range to summarize with mean/std like a continuous variable
Occupation	Categorical	Nominal	11 unordered job titles
Sleep Duration	Numeric	Continuous	Measured in hours to one decimal
Quality of Sleep	Categorical (Numeric rating)	Ordinal	1-9 self-report scale — order matters, but the gap between 6 and 7 isn't guaranteed equal to the gap between 7 and 8
Physical Activity Level	Numeric	Continuous	Minutes/day, effectively continuous
Stress Level	Categorical (Numeric rating)	Ordinal	Same reasoning as Quality of Sleep — a bounded self-report scale
BMI Category	Categorical	Ordinal	Normal < Overweight < Obese has a real clinical order, even though it's stored as text
Blood Pressure	Numeric	Continuous	Stored as "systolic/diastolic" — not usable numerically until parsed into two columns.
Heart Rate	Numeric	discrete (integer bpm)	
Daily Steps	Numeric	discrete (integer count)	
Sleep Disorder	Categorical	Nominal	None / Insomnia / Sleep Apnea — no inherent order

Typical Amount (Minutes) of Physical Activity

Median 60 min/day • roughly symmetric (skew = 0.07)

- Distribution is bimodal: a large cluster at 60–70 min/day and a smaller cluster at 30–45 min/day
- Overweight group shows the lowest median activity (45 min) vs. 60 min for both Normal and Obese groups
- Caveat: Obese subgroup is small (n=10) – treat the BMI comparison as directional, not conclusive



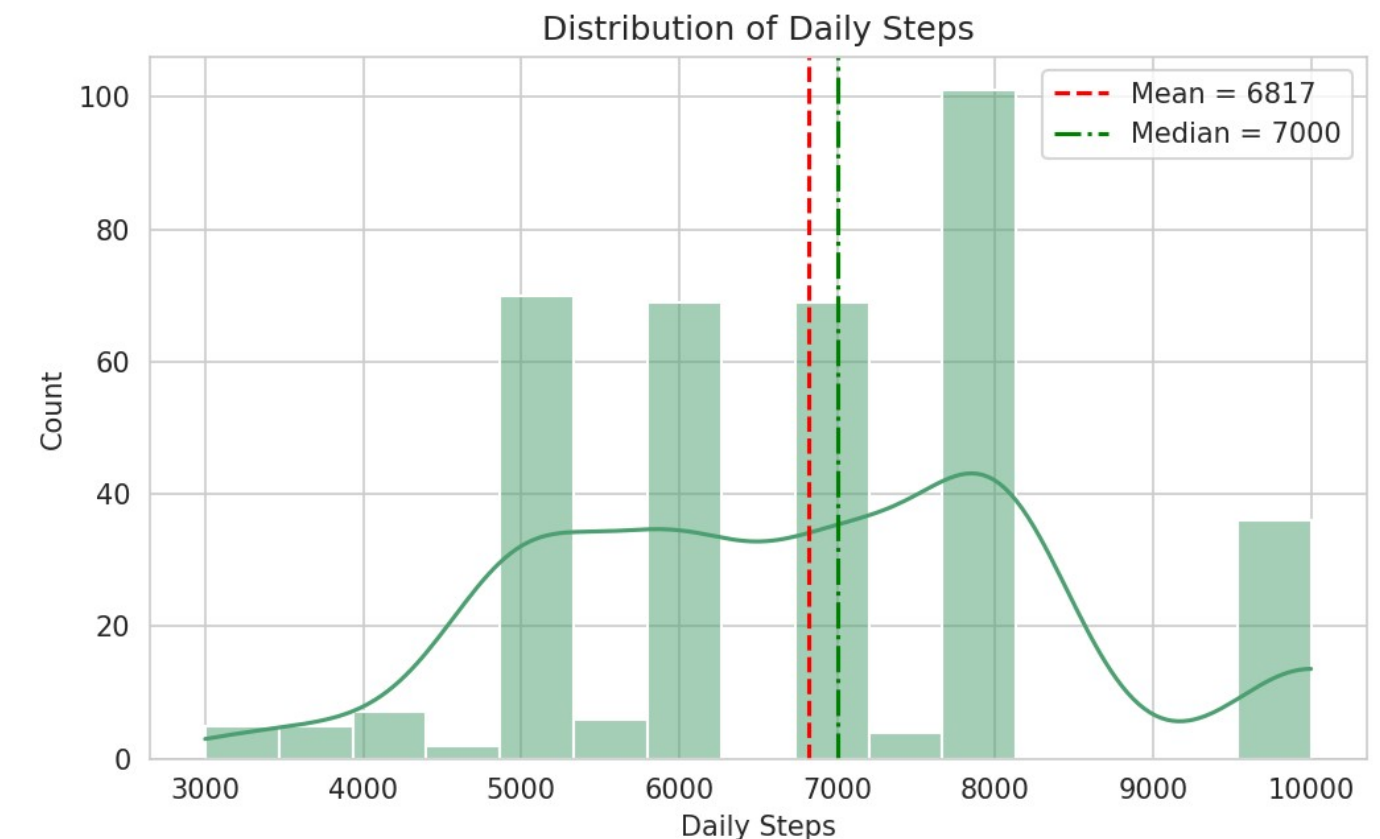
Analysis of Daily Steps Taken

Median ~7,000 steps/day • mild right skew (0.18)

- Mean 6,817 steps/day, std dev 1,618
- Wide spread from 3,000 to 10,000 steps
- Step counts vary noticeably across occupations
- Caveat: several occupations have very small samples (e.g. n=4 Software Engineers, n=1 Manager) – not a reliable ranking of which job is “healthiest”

Range of Daily Steps

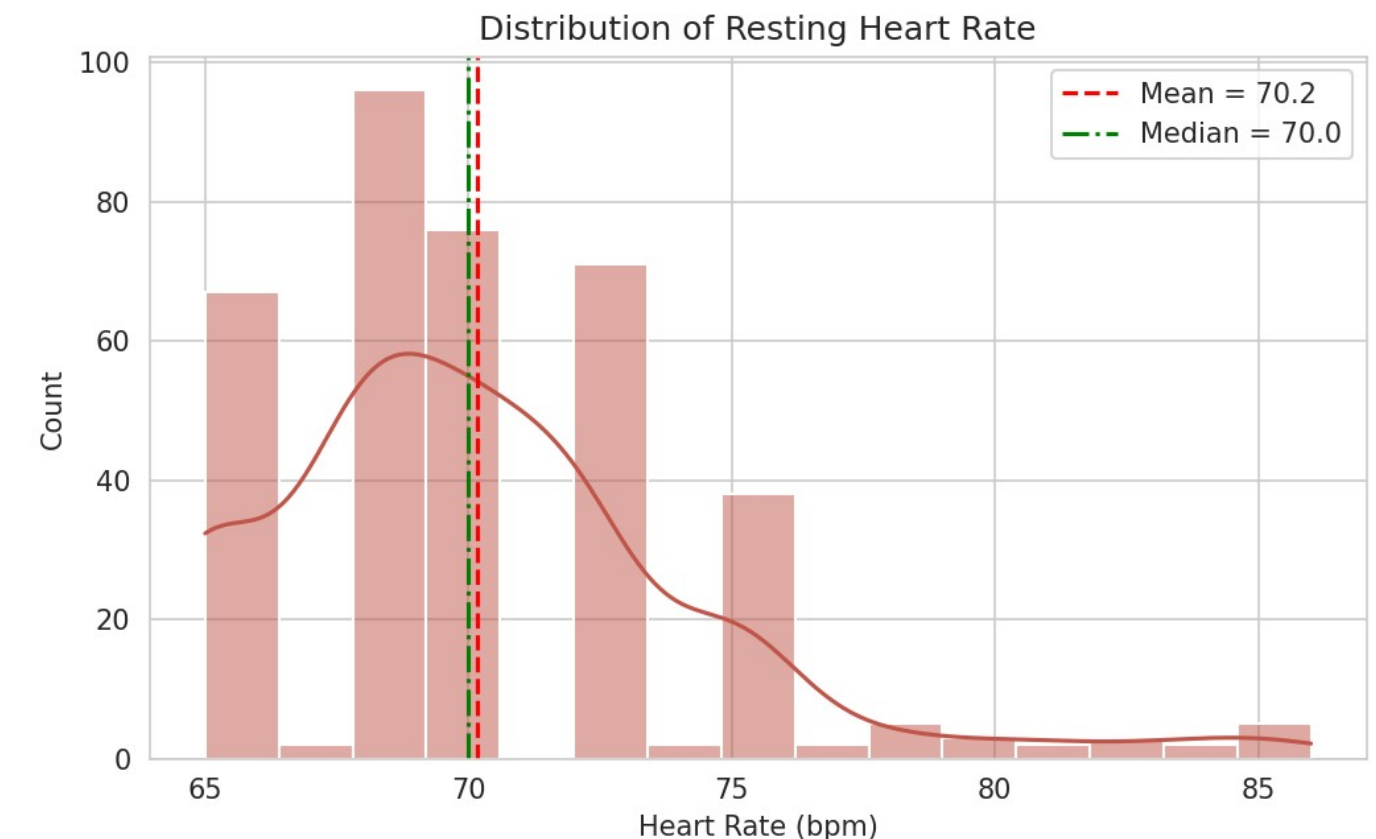
Min = 3000, Max = 10,000,
Range = Max - Min = 7000



Distribution of Heart Rates

Median 70 bpm • most skewed variable in the dataset (skew = 1.23)

- Right-skewed distribution: most respondents cluster in the low-to-mid 60s–70s bpm with a long tail of higher readings
- Because it's skewed, median (not mean) is the correct “typical value” to report
- Sleep Apnea (median 75 bpm) and Insomnia (median 72 bpm) groups run higher than the no-disorder group (median 70 bpm) – consistent with known links between sleep disorders and elevated resting heart rate



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